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Enjoyment of Life Predicts Reduced Type 2 Diabetes Incidence Over 12 Years

Findings from the English Longitudinal Study of Ageing (ELSA)

Panagi L, Hackett RA, Steptoe A, Poole L

ST6012/ST2403: Survival Analysis

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Introduction & Aim of the Paper

Research Question:

Does enjoyment of life independently predict incident type 2 diabetes over 12 years in adults aged 50+, after adjustment for sociodemographic, behavioural and clinical factors?

The Context

T2D is a major and growing public health burden among older adults in England. Lifestyle and biological risk factors are well-established. Psychosocial predictors, especially positive ones, are far less studied in prospective designs.

The Gap

Most studies ask whether depression raises T2D risk. This paper asks the reverse. It also separates enjoyment from purpose — something prior studies rarely did.

The Aim

To test whether enjoyment of life predicts incident T2D over 12 years after full adjustment, and to estimate how much of any association is explained by measured covariates.

Background

What we already know about well-being & health

Positive affect is consistently associated with lower CVD, stroke, and all-cause mortality across multiple large prospective cohorts.

Hedonic enjoyment (liking what you do) is conceptually and empirically distinct from eudaimonic well-being (sense of purpose). They measure different things.

Cross-sectional studies show T2D patients report lower well-being, but cross-sectional data cannot establish which came first.

The authors argument: given the one-directional limitation, you must study enjoyment disease prospectively to make any progress.

The proposed biological & behavioural pathways to T2D

Low enjoyment → chronic low-grade psychological stress → dysregulated HPA axis → elevated cortisol → insulin resistance → T2D.

Behavioural route: less enjoyment → less physical activity, poorer dietary choices, higher smoking rates → metabolic risk accumulates.

Inflammatory route: sustained negative affect elevates IL-6 and CRP, both impair insulin signalling and glucose metabolism.

Critical caveat: none of these pathways are directly measured in ELSA. These are proposed mechanisms, not verified ones.

Methods - Study Design

Source Population	Exclusions Applied	Exposure	Outcome
ELSA: national probability sample Adults aged ≥ 50 , England Baseline: Wave 2 (2004–05) Starting N = 8,780	Prevalent T2D at baseline Missing CASP items Missing covariates (HbA1c largest, n=2,243) Died before first follow-up → Final N $\approx$ 4,134	4-item CASP-19 subscale Scored 4–16 (higher = more enjoyment) Baseline Wave 2 ONLY  Single snapshot — 12 yrs	Incident T2D — new cases only Self-reported physician diagnosis at each wave NOT nurse HbA1c Followed to Wave 8 (2016–17)

Timeline: Wave 2 (2004–05) → W3 → W4 → W5 → W6 → W7 → Wave 8 (2016–17) | Average 12-year follow-up

Covariates adjusted for:

. Age · Sex · Ethnicity · Wealth quintile · Marital status · Smoking · Alcohol · Physical activity · BMI · Hypertension · CHD · HbA1c (baseline nurse-measured) | Depression: secondary analysis only

Methods - Statistical Analysis

1

Preliminary Analysis

Pearson r for continuous variables. Independent samples t-tests and Kruskal-Wallis H tests for categorical variables. Used to examine associations between well-being scores and sample characteristics at baseline.

2

Cox Proportional Hazards Regression

Five sequential models for each exposure separately. Exposure entered as continuous per 1-unit increase. Time to event in months from Wave 2 to first self-reported T2D diagnosis. Proportional hazards assumption tested via log minus log survival plots - not violated.

3

PPAE - Formal Mediation Estimate

Percentage of Protective Association Explained (Lin et al. 1997). Formally estimates what proportion of the exposure-outcome association is accounted for by each covariate group. This is a published quantitative method, not informal HR attenuation. Results: sociodemographic 27%, behavioural 27%, clinical 18%, all covariates 36%.

4

Secondary Analyses and Sensitivity Checks

Excluded T2D cases within the first 2 years to test reverse causation (n=4,064). Separately excluded participants with HbA1c 48 or above at baseline (n=4,068). Tested age and sex interaction terms, both non-significant. Main finding held in all three checks.

Results - Baseline Characteristics (Table 1, N = 4,134)

Overall sample at baseline, Wave 2 (2004/2005). Values are n (%) or Mean plus or minus SD. Both association columns reproduced faithfully from Table 1.

Characteristic	n (%) or Mean $\pm$ SD	Assoc. with Enjoyment of effect size, p value	Assoc. with Purpose in Life effect size, p value
CASP enjoyment of life score	14.23 $\pm$ 1.75	<i>Primary exposure</i>	r = 0.62, p < 0.001
CASP purpose in life score	3.59 $\pm$ 0.71	r = 0.62, p < 0.001	<i>Primary exposure</i>
Age (years)	64.97 $\pm$ 8.99	r = -0.05, p = 0.003	r = -0.03, p = 0.040
Sex (% women)	2305 (55.8%)	d = 0.07, p = 0.026	d = 0.04, p = 0.231
BMI (kg/m ²)	27.56 $\pm$ 4.63	r = -0.03, p = 0.043	r = 0.01, p = 0.409
HbA1c (mmol/mol)	36.2 $\pm$ 2.46	r = -0.05, p = 0.004	r = -0.04, p = 0.007
Depression cases (% yes)	526 (12.7%)	d = 0.93, p < 0.001	d = 0.77, p < 0.001
Smoking (% smokers)	564 (13.6%)	d = 0.33, p < 0.001	d = 0.21, p < 0.001
Physical activity > 1x/week	2463 (59.6%)	e2 = 0.03, p < 0.001	e2 = 0.01, p < 0.001
Hypertension cases (% yes)	1698 (41.1%)	d = 0.11, p < 0.001	d = 0.01, p = 0.735
CHD cases (% yes)	400 (9.7%)	d = 0.21, p < 0.001	d = 0.11, p = 0.035

Source: Table 1, Panagi et al. JECH 2021. Both association columns reproduced verbatim. Our primary exposure is enjoyment of life (third column).
d = Cohen's d, r = Pearson r, e2 = epsilon-squared. Yellow rows = key variables (exposure, HbA1c, depression).

Results - Cox Regression: Sequential Models (Table 2, 4 N = 4,134)

Enjoyment of Life predicting T2D incidence (per 1-unit increase)

	HR	95% CI	P value	PPAE
Unadjusted model	0.89	0.84 to 0.94	<0.001	
+Sociodemographic variables	0.92	0.87 to 0.97	0.003	27%
+Behavioral variables	0.92	0.87 to 0.97	0.004	27%
+Clinical variables	0.91	0.86 to 0.97	0.003	18%
+All variables	0.93	0.87 to 0.99	0.020	36%

Purpose in Life predicting T2D incidence (per 1-unit increase)

	HR	95% CI	P value	PPAE
Unadjusted model	0.85	0.74 to 0.99	0.032	
+Sociodemographic variables	0.90	0.77 to 1.04	0.154	33%
+Behavioral variables	0.87	0.75 to 1.01	0.066	13%
+Clinical variables	0.93	0.80 to 1.09	0.389	53%
+All variables	0.92	0.79 to 1.08	0.322	47%

Key contrast: Enjoyment of life significant across all 5 models (final HR = 0.93, p = 0.020). Purpose in life loses significance once covariates are added (final HR = 0.92, p = 0.322). Sociodemographic and behavioural factors each explain 27% of the enjoyment association.

Source: Table 2,4 Panagi et al. JECH 2021. CI, confidence interval; HR, hazards ratio; N, number; PPAE, percentage of protective association explained; T2D, type 2 diabetes.

Results - Fully Adjusted Model (Table 3, N = 4,134)

All predictors in the fully adjusted Cox model — enjoyment of life predicting T2D incidence from 2006/07 to 2016/17

Variable	HR	95% CI	P value
CASP enjoyment of life score	0.93	0.87 to 0.99	0.020
Age (years)	1.01	1.00 to 1.03	0.070
Sex (reference: men)	0.64	0.50 to 0.81	< 0.001
Financial wealth — Quintile 1	1	0.67 to 1.52	0.962
Financial wealth — Quintile 2	1.01	0.79 to 1.65	0.472
Financial wealth — Quintile 3	1.14	0.67 to 1.42	0.891
Financial wealth — Quintile 4	0.97	0.46 to 1.04	0.076
Financial wealth — Quintile 5	0.69		
Ethnicity (reference: white)	1.87	0.83 to 4.23	0.131
BMI (kg/m ²)	1.10	1.08 to 1.13	< 0.001
Hypertension (reference: no)	1.35	1.07 to 1.71	0.013
HbA1c (%)	2.58	2.33 to 2.85	< 0.001

Primary result

HR 0.93 per 1-unit increase in enjoyment.
7% reduction in T2D hazard. $p = 0.020$.

Strongest predictor

HbA1c: HR 2.58 (2.33 to 2.85), $p < 0.001$.
Every 1% increase raises T2D hazard more than 2.5-fold.

Sex effect

Women have 36% lower hazard than men (HR 0.64).
No significant sex x enjoyment interaction.

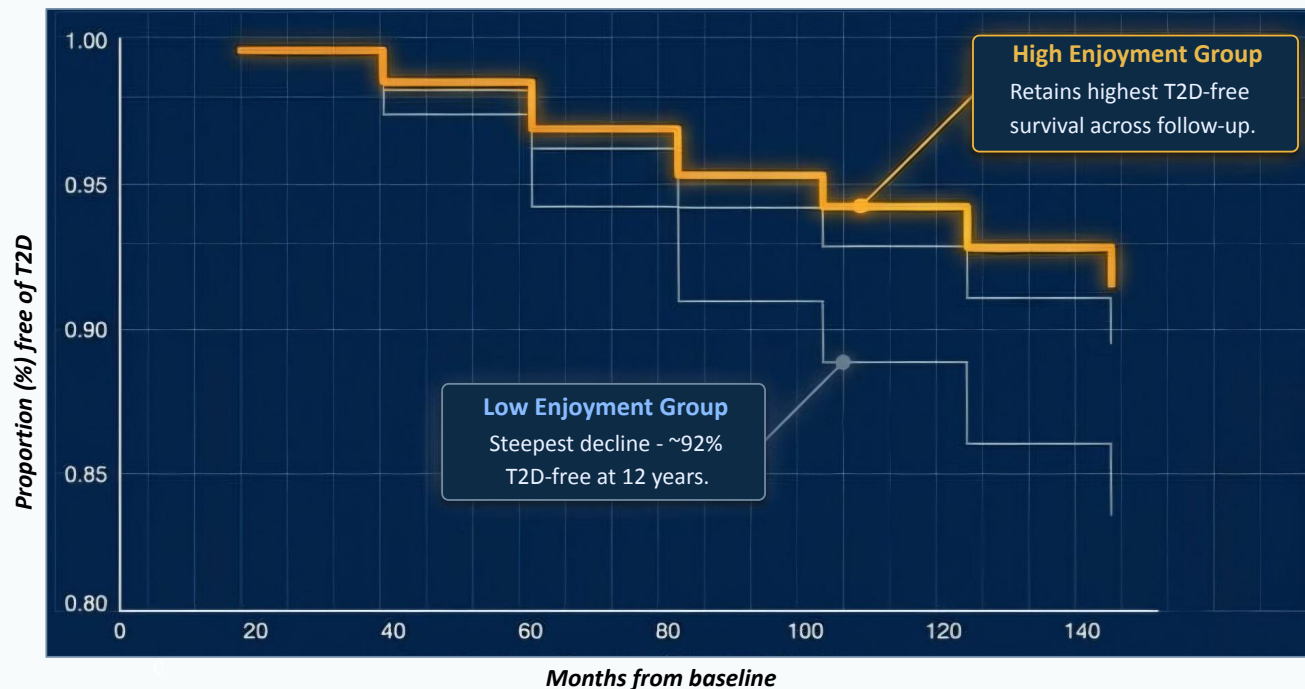
Lifestyle not significant

Physical activity, smoking and alcohol are not significant in the fully adjusted model ($p > 0.05$).

Source: Table 3, Panagi et al. JECH 2021. CI, confidence interval; HR, hazards ratio; T2D, type 2 diabetes. Yellow rows = primary exposure (enjoyment) and strongest predictor (HbA1c). 312 incident T2D cases (7.5%) over average 11.6-year follow-up.

Results - Kaplan-Meier Survival Curves (Figure 2)

Proportion (%) free of T2D by baseline enjoyment of life category. Three groups. Adjusted for age and sex. Source: Figure 2, Panagi et al. JECH 2021.



CASP-19 Enjoyment Groups

- High (score 15–16)
- Middle (score 13–14)
- Low (score 4–12)

Key Statistics

312 incident T2D cases
7.5% of analytical sample
Average follow-up: 12 years

Unadjusted HR (per 1-unit increase in enjoyment):
HR = 0.89 (0.84 to 0.94)
 $p < 0.001$

Middle group sits consistently between high and low throughout.

Source: Figure 2, Panagi et al. JECH 2021. ⚠ Adjusted for age and sex only — not the fully adjusted results.

Discussion

Every 1-unit increase in enjoyment of life was associated with a 7% reduction in T2D hazard over 12 years (HR = 0.93, 95% CI 0.87 to 0.99, $p = 0.020$), after adjusting for sociodemographic, behavioural and clinical factors.

The PPAE finding: partial but not full explanation

36% of the association explained by measured covariates. Two thirds remains unexplained. Biological pathway proposed but not directly tested in ELSA.

Purpose in life - a null result that matters

Fully adjusted HR = 0.92, $p = 0.322$. Not a failed replication. Hedonic and eudaimonic well-being are different things and behave differently here.

Independence from depression - and why it matters

Secondary analysis HR = 0.91, $p = 0.002$. The association holds independently of depression. Not simply a proxy for not being depressed.

No age or sex moderation

Both interaction terms non-significant. The protective effect holds across all age groups and both sexes.

Secondary analysis HR for enjoyment independent of depression: 0.91 (0.86 to 0.97), $p = 0.002$. Purpose in life fully adjusted HR: 0.92 (0.79 to 1.08), $p = 0.322$.

Critical Appraisal - Strengths

1 Large, nationally representative sample

N = 4,134. Probability-based national sample of English adults aged 50 and over. Average follow-up 12 years.

2 Prospective design with long follow-up

Well-being measured at Wave 2 before any T2D developed. Follow-up to Wave 8 across seven biennial waves.

3 Objective baseline clinical measures

HbA1c and BMI measured by nurses during home visits. Stronger than self-report. Reduces misclassification at baseline.

4 Reverse causality addressed by two methods

Excluded T2D cases within 2 years of baseline. Also excluded HbA1c 48+ at baseline. Two complementary checks.

5 Separates hedonic from eudaimonic well-being

Enjoyment and purpose tested independently. The contrasting results would have been invisible in any combined measure.

6 Depression sensitivity analysis conducted

Association tested with and without depression in the model. Effect is distinct from the mere absence of depression.

Critical Appraisal - Limitations

1 Complete case analysis — 4,646 participants excluded

Just over half the Wave 2 sample excluded. Included group was younger, wealthier and healthier. Systematic selection bias.

2 Self-reported T2D outcome

Based on self-reported physician diagnosis, not clinical records or HbA1c at follow-up. Undiagnosed cases missed entirely.

3 Well-being measured at a single time point only

Measured once at Wave 2. Treated as fixed for 12 years. Well-being changes. One snapshot cannot capture a decade.

4 Proposed biological mechanisms not measured

Cortisol and inflammatory pathways proposed from other studies. Not measured in ELSA. The mechanism is an inference, not a finding.

5 PPAE cannot distinguish mediation from confounding

PPAE estimates contribution but cannot separate a mediator from a confounder.

6 Limited ethnic diversity and missing diet covariate

99% white sample. Diet not included as a covariate despite being a plausible pathway. Limits generalisability.

Red = critical limitations. Yellow = moderate limitations. Grey = acknowledged by the authors themselves.